

Crossroad Logic Simulation and PCB Implementation

Project Proposal

Course: Formal Models

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Abstract

This project is about making a digital logic system that controls how traffic lights work at a crossroad. We are taking the project from a basic idea to a real physical board.

Traffic management at intersections requires strict, fail-safe logical sequences to prevent accidents and optimize traffic flow. Designing these sequences falls under the domain of formal logic and digital circuit design.

This topic is relevant to Formal Models because we are using Boolean logic and state transitions to solve a real problem. It shows how the logic we learn in class can be used to build actual hardware.

Problem statement

While the theoretical aspects of formal logic and state machines are thoroughly covered in academic settings, there is often a gap in translating these abstract models into tangible, physical engineering solutions. This project addresses that gap by taking a purely logical construct (the rules governing a traffic intersection) and proving its validity through both software simulation and physical hardware fabrication.

Objectives

- To accurately define the formal logical rules and state transitions for a standard crossroad traffic light system.
- To derive the necessary logic gate expressions (e.g., AND, OR) required to execute these state transitions.
- To achieve a fully verified simulation of the digital logic circuit using Proteus 8.
- To successfully demonstrate the physical application of formal models by manufacturing a functioning Printed Circuit Board (PCB) containing the sealed components.

Methodology

- The project will be carried out in three main phases: theoretical design, software simulation, and hardware implementation. First, truth tables and state machines will be developed to define the traffic logic. Next, these expressions will be mapped to standard logic gates.
- Iterative testing will be conducted in a simulated environment to verify the logic before moving to the fabrication phase. The final phase involves routing the PCB, printing it, and soldering the active and passive components.
- The primary software tool used will be Proteus 8 for both the logic circuit simulation and the PCB layout design. Hardware components will include discrete logic ICs (or transistor-based logic equivalents), LEDs, resistors, and physical PCB fabrication materials.

Expected Results

- At the end of the project, the team will deliver a fully functional, verified Proteus 8 simulation file demonstrating the correct traffic sequences.
- Additionally, the primary physical deliverable will be a custom-printed, fully soldered PCB that actively physically executes the designed logic to control miniature traffic lights.

Additional Information

Sources

- We will use datasheets for the logic gates and components to make sure we connect everything correctly, alongside standard traffic light timing and sequence guidelines.

Team Roles

- **Emin Hadžiabdić** (ehadziabdi1@etf.unsa.ba)
 - ◆ Primary lead for the software phase, responsible for building and verifying the logic circuit simulation within Proteus 8. Additionally tasked with translating the simulated circuit into a physical PCB layout (routing) and assisting with the final hardware assembly and soldering.
- **Muhammed Pašić** (mpasic4@etf.unsa.ba)
 - ◆ Lead for the hardware fabrication phase. Primarily responsible for the physical assembly, component placement, and precise soldering of the active and passive elements onto the printed circuit board to ensure a functional final prototype.
- **Armin Memišević** (amemisevic2@etf.unsa.ba)
 - ◆ Lead for theoretical logic design and implementation. Responsible for deriving the necessary Boolean expressions and state machines, as well as providing technical assistance in translating this abstract logic into the practical Proteus software environment.

Note: While we have assigned these primary leads to meet the project requirements, the team plans to work closely together on every phase of the project to ensure full collaboration.